Medicaid Work Requirements and Hospital Readmissions: A Difference-in-Differences Analysis in Arkansas

Alexander Klautky, Katelyn Kim, Anika Kaur

In recent years, several US states have implemented work requirements for Medicaid eligibility to increase employment in low-income populations. Arkansas was the first state to fully implement such a policy in 2018, resulting in the disenrollment of 18,000 individuals before it was blocked in mid-2019. With states like Ohio now seeking to enact a similar policy, there is growing concern about the potential negative health consequences. This paper evaluates the nature of these consequences by analyzing the effects of the policy on excess readmission ratios (ERR), a key hospital performance metric created and used by the Center for Medicare and Medicaid Services (CMS). Using ERR data from 2014 to 2024, we leveraged a Difference in Differences (DiD) regression framework to compare ERR performance between Arkansas and three control states. However, the effects we observed were small in magnitude and statistically insignificant. These findings suggest that work requirements may affect healthcare quality despite the limited effects observed in our analysis.

Introduction

Since 2010, Medicaid work requirement policy has become increasingly popular among Republican states, including Ohio. As of 2025, Arkansas is the first and only state to have fully implemented Medicaid work requirements. During its brief rollout, starting in June 2018, it was reported that approximately 18,000 individuals lost Medicaid coverage before a federal judge blocked this policy in March 2019 (Skinner, 2024). Ohio received approval for similar work requirements in 2019 under the Trump administration, but implementation was paused due to the COVID-19 pandemic. Before Ohio republicans could resume their pursuit of the policy, the Biden Administration revoked the approval in 2021. Now, under the new Trump administration, Ohio has resubmitted a proposal to the Centers for Medicare and Medicaid Services (CMS) to implement work requirements (Kasler, 2025).

As this proposal re-emerges, it is important to understand the effects this policy may have on the health outcomes of Ohio's Medicaid population as it risks revoking coverage, disrupting access to necessary care for vulnerable groups. Currently, eligibility for Ohio Medicaid is determined by an individual's income relative to the Federal Poverty Level (FPL). Following Medicaid expansion under the Affordable Care Act in 2014, non-elderly adults between the ages of 19 and 64 with incomes up to 138% of the FPL also became eligible (McGee, Akah, & Wiselogel, 2025). Under the work requirements proposal, these non-elderly adults would have to meet certain conditions in order to remain eligible for Medicaid. One such condition requires individuals to work or engage in community service for a minimum of 20 hours a week to retain Medicaid coverage. While the proponents of this policy believe that the requirements would help reduce federal spending, boost employment, and help individuals become more independent, evidence shows that many Medicaid recipients are already working part-time or face significant barriers to stable employment (Garfield, Rudowitz, Musumeci, & Damico, 2018). Proponents of the policy also claim that there is a connection between employment and better health. However, insufficient literature casts doubt on whether this connection is actually significant.

Given that insurance coverage plays a key role in readmission rates (Bailey, Weiss, Barrett, & Jiang, 2019), we will theoretically predict that a population shift from Medicaid to uninsured coverage should cause the ERR to increase (Wier, 2011). Because of this, we will interpret an increase in ERR as an indication that the distribution of Medicaid and uninsured individuals has changed. Therefore, our hypothesis will be one-tailed to capture the expected shift in ERR.

H_0: The Arkansas implementation of Medicaid work requirements has a negative or neutral impact on expected readmission ratio (ERR) outcomes.

H_1: The Arkansas implementation of Medicaid work requirements has a positive impact on expected readmission ratio (ERR) outcomes.

Literature Review

Much of the existing literature on Medicaid work requirements and Medicaid expansion has focused on the consequences that coverage loss has on healthcare access and affordability. Evidence from Sommers, Chen, Blendon, Orav, and Epstein (2020) highlights how Arkansas's 2018 implementation resulted in widespread disenrollment and adverse health-related behaviors. Through a series of random digit dialing telephone surveys in 2016, 2018, and 2019 of low-income adults, they found that among those who were affected, 50% reported serious problems with medical debt, 56% had delayed care due to costs, and 64% delayed taking medications (Sommers et al., 2020). Along the same lines, Garfield et al. (2018) and Katch, Wagner, and Aron-Dine (2018) argue that coverage disruptions disproportionately affect already vulnerable populations, like individuals with mental illness, chronic conditions, or fluctuating work hours. These vulnerable populations already face barriers to entering the workforce and are predisposed to worse health outcomes, so the loss of coverage would likely exacerbate this effect. In addition, Jung and Shrestha (2024) simulate the effects of Medicaid work requirements and find that although there is a slight rise in labor force participation, Medicaid disenrollment is primarily concentrated among high-risk, low-income individuals. They suggest that the administrative costs and coverage losses outweigh any marginal gains from employment.

Instead of looking at healthcare accessibility, we have decided to examine healthcare quality by utilizing Expected Readmission Ratios (ERR) to estimate the direct health effects of the policy in Arkansas. ERR is a metric developed by the CMS that is often used to assess how well hospitals manage acute conditions and ensure appropriate follow-up care. Although Zuckerman, Sheingold, Orav, Ruhter, and Epstein (2016) does not directly examine insurance loss, their study supports the use of ERR as a valid quality indicator because it demonstrates that policy interventions can have an impact on ERR. While there are few studies that have relied on ERR to examine the effects of Medicaid work requirements, we were able to create a framework for our analysis by utilizing ERR outcomes in Arkansas to offer a new perspective on the medical impacts of this policy. This not only reinforces concerns about coverage loss but also provides more precise insight into how such policy changes may undermine healthcare quality. Since Ohio is proposing the implementation of work requirements, we aim to understand whether such policies are likely to fulfill their intended goals or instead, lead to worse health outcomes for those most in need.

Data

This analysis uses emergency room Expected Readmission Ratios (ERR) from the publicly available Center for Medicare and Medicaid Services (CMS) database to evaluate the effects of Medicaid coverage loss on healthcare quality. ERR reflects the ratio of predicted to expected readmission rates for medical conditions requiring immediate attention. The CMS uses ERR rates in every state as a performance metric to assess a hospital's ability to treat acute medical conditions, determining that hospital's reimbursement. ERR is particularly sensitive to certain conditions such as heart failure and pneumonia, which are highly inelastic to price changes, unlike elective procedures like joint replacements. ERR tends to increase when a larger proportion of the population lacks insurance, making it a fitting measure of differences in outcomes between insured and uninsured populations (Bailey et al., 2019). While there are some concerns regarding the variability of ERR, these issues are discussed further in the limitations section. The control states, Mississippi, Missouri, and Oklahoma were selected by two criteria: 1. A control state must be similar to the treated state in terms of its population's healthcare consumption patterns. 2. A control state should not have a partial intervention of the same type as the treated state. Partial intervention states like Georgia are particularly poor candidates for control states because they would confound estimates of the treatment effect and bias it towards zero.

Our final data set, derived from the raw CMS data set, consists of 6,632 observations distributed across 280 hospitals and three of the six ERR conditions. We are using a selected set of ERR conditions for each hospital in the treatment state and in the control states. We excluded three conditions: Hip and Knee Replacement (THA), Chronic Obstructive Pulmonary Disease (COPD), and Coronary Artery Bypass Graft (CAPG) because they were not consistently present in the pretreatment years. We used the following conditions: Heart Failure (HF), Pneumonia, and Acute Myocardial Infarction (AMI). Each observation is at the hospital-condition level. The sample is also restricted between the years 2014 to 2024, giving us four years before the policy implementation to support a robust testing of the parallel trends assumption, and four years after to evaluate both immediate and longer-term effects. The year 2023 was excluded because it did not report Pneumonia.

Methodology

To estimate the impact of the policy on Arkansas ERR rates, we chose to employ a few variations on the difference-in-differences (DiD) framework. This allows us to assess whether a treatment effect exists, when it emerges, and evaluate the parallel trends assumption. The DiD framework assumes that in the absence of the treatment, treated and controlled hospitals within states should experience parallel trends in ERR outcomes. We can test this by evaluating the magnitude and significance of the difference between Arkansas and control before treatment.

We based our analysis on the generic DiD framework, using a regression model to estimate the effects of the average treatment effect on ERR. The canonical DiD regression is given by the equation:

$$Y_{it} = \beta_0 + \beta_1 \cdot \text{Post}_t + \beta_2 \cdot \text{Treatment}_i + \beta_3 \cdot (\text{Post}_t \times \text{Treatment}_i) + \varepsilon_{it}$$

We modified the basic specification to include year indicators, interactions between treatment status and each year, and condition-fixed effects. This specification is given by the following equation:

$$\text{Outcome}_{itc} = \beta_0 + \sum_{t \neq 2017} \beta_t \text{Year}_{t,it} + \gamma \text{Intervention}_i + \sum_{t \neq 2017} \delta_t \left(\text{Year}_{t,it} \times \text{Intervention}_i\right) + \sum_c \theta_c \ \text{Cond}_c + \varepsilon_{itc}$$

Where:

- Outcome_{itc}: the ERR for hospital i, in year t, for condition c.
- Intervention_i: a dummy equal to 1 if hospital i is in Arkansas.
- $\text{Year}_{t,it}$: a set of dummies for each year from 2014 to 2024

- $\text{Year}_{t,it} \times \text{Intervention}_i$: the DiD interaction terms for each year k.
- δ_k: the dynamic treatment effect over time (main coefficient of interest).
- Cond_c : the dummy for diagnostic condition (AMI, HF, or PN).
- ε_{itc} : clustered error term at the hospital level (PROV).

Several key treatment-related variables were generated. *Intervention* (1 if ARK, 0 otherwise) indicates the policy receiving group (ARK). *Post* (1 for years $\geq$ 2017 and 0 otherwise) denotes the pre and post-treatment periods that represent the implementation of the 2018 policy. *rel_year_shifted* indicates all years with respect to their distance from the baseline year 2017 so that the first year 2014 corresponds with one and the baseline year corresponds with five. *Cond_id* is a cleaned and standardized representation of the three selected ERR conditions, which is used as a categorical variable to generate the dummy variables. All standard errors are clustered at the hospital level to account for repeated observations of the same provider (hospital) over time. A year control variable like *i.year* was not used because the *ib2017year* variable already controls for year effects. We also conducted a robustness check by modifying the first specification to group the pre and post-years together to estimate the aggregated effects on the years before and after the policy implementation.

Results

We used the regression specification given above to estimate the impacts of the Arkansas policy implementation on ERR outcomes. We had two goals in mind for this analysis: first, to validate the parallel trends assumption; second, to determine the magnitude and significance of the post-treatment effects of the intervention policy on Arkansas. The results from this regression are displayed in Figure 1 and Table 1 below.

The joint F-test for significance across all 21 interaction terms yields (21, 289) = 1.55 with a p-value of (p = 0.0598). This indicates a marginal joint significance at the 10% level, suggesting that the treatment had a potentially nonrandom effect on ERR outcomes. However, the low R-squared (0.0139) and adjusted R-squared (0.0108) indicate that the explained variance in ERR is limited.

To test the parallel trends assumption, we look at the pre-treatment coefficients (years 2014 through 2017). These interactions range from -0.0074 to 0.0008 and are statistically insignificant (p = 0.529 to 0.883), indicating no relevant difference between Arkansas and the control states before the 2018 intervention, confirming the parallel trends assumption. Additionally, we visually confirmed this result by observing Figure 1. This supports the validity of the DiD strategy to test our hypothesis.

The coefficients of the post-treatment interaction terms (years 2018-2024) are consistently negative after the policy, indicating that the ERR outcomes consistently decreased relative to the control states. The largest effects occur in 2022 (-0.0155, p=0.137) and 2024 (-0.0143, p=0.12). However, none of these coefficients are statistically significant as their p-values exceed the predetermined 0.05 alpha level. We confirmed these results with a robustness check using a grouped-year regression where all pre and post-years were collapsed into one binary indicator, see table 2. This supports the parallel trends assumption by showing that the interaction term for the pre-policy period was small (–0.0017) and insignificant (p=0.456). We also confirmed that the post-intervention coefficient was small (0.0049) and insignificant (p=0.528). Both analyses show that the drop in the post-treatment year of the treated group is small, negative, insignificant, and imprecise.

Figure 1: Outcome by Treatment Status Over Time.

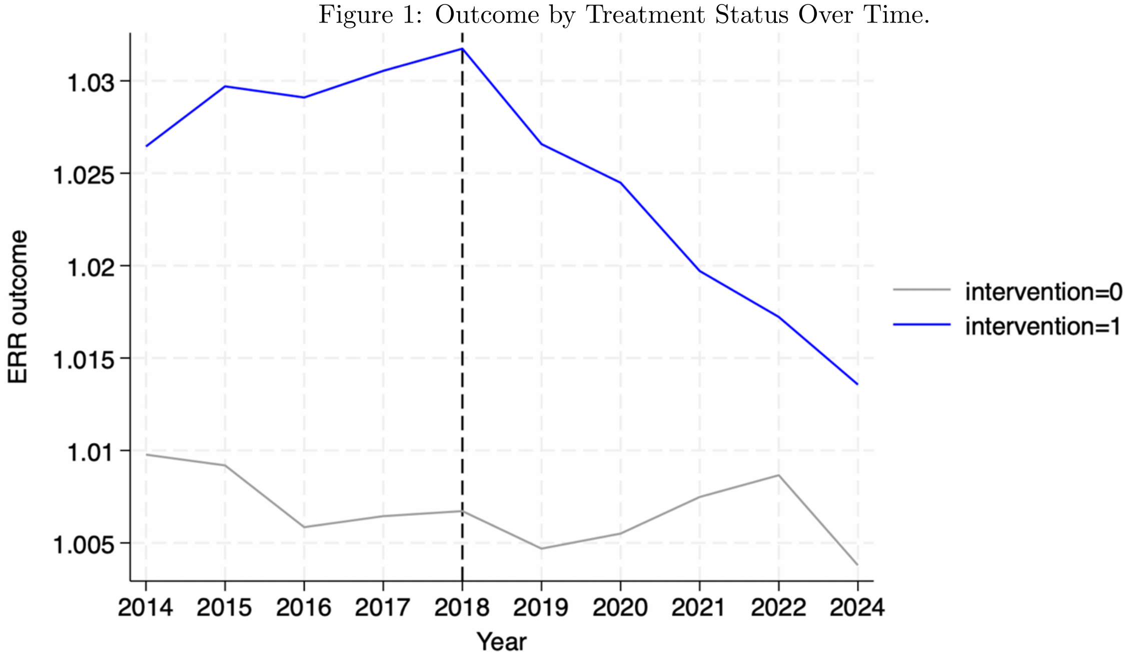

Figure 11 displays predicted ERR in Arkansas vs. control states (MO, OK, MS) from 2014–2024.

Table 1: Event Study Regression Results (Excluding t-values)

Outcome	Coefficient	Std. Err.	P > $|t|$
year			
2014	0.0033275	0.0042993	0.440
2015	0.0027443	0.0038018	0.471
2016	-0.0005981	0.0024771	0.809
2018	0.0002726	0.0025209	0.914
2019	-0.0017578	0.0026941	0.515
2020	-0.0009446	0.0030405	0.756
2021	0.0010364	0.0031083	0.739
2022	0.0022149	0.0029928	0.460
2024	-0.0026578	0.0032799	0.418
intervention			
1	0.0240932	0.0091345	0.009
year x intervention			
2014	-0.0074185	0.0114768	0.519
2015	-0.0035844	0.0104676	0.732
2016	-0.0008461	0.00573	0.883
2018	0.0009251	0.0053246	0.862
2019	-0.0022048	0.007542	0.770
2020	-0.0051107	0.0088851	0.566
2021	-0.0118655	0.0099435	0.234
2022	-0.0155323	0.0104159	0.137
2024	-0.0143167	0.009109	0.120
condition			
HF	0.0050948	0.0028262	0.072
PN	0.0013424	0.0028004	0.632
_cons	1.004178	0.0033659	0.000

Table 1 displays regression output for our DiD models.

Table 2: Grouped-Year Regression Results

Variable	Coefficient	Std. Err.	P > $|t|$
post			
1	-0.0016972	0.0022758	0.456
intervention			
1	0.0210892	0.0086651	0.016
post x intervention			
1 1	-0.0049823	0.0078931	0.528
condition			
HF	0.0050614	0.0028027	0.072
PN	0.0012921	0.0027927	0.644
_cons	1.005588	0.0036617	0.000

Conclusion

Based on our theoretical expectations, we hypothesized that the ERR for Arkansas would increase after the implementation of the work requirements policy. However, our difference in differences (DiD) analysis found that Arkansas' ERR consistently decreased relative to the control states. While these findings were small and statistically insignificant, they still suggest an effect opposite to our expectations. Therefore, we fail to reject the null hypothesis that the Arkansas implementation of Medicaid work requirements has a negative or neutral impact on expected readmission ratio (ERR) outcomes.

A possible explanation for our results is that the implementation period of the policy (2018-2019) was too short for a trend to emerge. Another possibility is that the effects of the COVID-19 pandemic, which emerged in 2019, positively influenced the ERR data, confounding our results. This "COVID effect" is also exacerbated by the small size of the impacted group, 18,000 individuals lost Medicaid coverage (Skinner, 2024). This is relatively small compared to the 1.2 million individuals estimated to be on Medicaid in Arkansas.

The short duration of the policy implementation, between 2018 and 2019, likely limited our ability to detect a significant effect in ERR. Additionally, the onset of the COVID-19 pandemic introduced potential confounding factors, particularly for delayed effects after 2019. Finally, due to the statistical insignificance and small magnitude of the observed effects, it remains unclear whether the slight decrease in Arkansas ERR reflects an improvement, a deterioration in Medicaid coverage, or simply random variation.

References

Bailey, M. K., Weiss, A. J., Barrett, M. L., & Jiang, J. J. (2019). *Characteristics of 30-Day All-Cause Hospital Readmissions, 2010-2016 #248.* Retrieved 2025-04-17, from `https://hcup-us.ahrq.gov/reports/statbriefs/sb248-Hospital-Readmissions-2010-2016.jsp`

Fingar, K. R., & Jiang, H. J. (n.d.). A Comparison of All-Cause 7-Day and 30-Day Readmissions, 2014.

Folse, M., Bridges, J., & DiGiorgio, A. (2024, September). "How Will Medicaid Work Requirements Affect American Healthcare? A Look at What Past and Present Policy Tells Us". *INQUIRY*, *61*, 00469580241251935. Retrieved 2025-04-17, from `https://doi.org/10.1177/00469580241251935` (Publisher: SAGE Publications Inc) doi: 10.1177/00469580241251935

Garfield, R., Rudowitz, R., Musumeci, M., & Damico, A. (2018). Implications of Work Requirements in Medicaid: What Does the Data Say?

Jung, J., & Shrestha, V. (2024, August). *Medicaid Work Requirements, Labor Market Effects and Welfare* [SSRN Scholarly Paper]. Rochester, NY: Social Science Research Network. Retrieved 2025-04-17, from `https://papers.ssrn.com/abstract=4928303`

Kasler, K. (2025, March). *Ohio once again wants to require people in Medicaid expansion to work or lose coverage.* Retrieved 2025-04-17, from `https://www.statenews.org/government-politics/2025-03-06/ohio-once-again-wants-to-require-people-in-medicaid-expansion-to-work-or-lose-coverage` (Section: Government/Politics)

Katch, H., Wagner, J., & Aron-Dine, A. (2018). *Taking Medicaid Coverage Away From People Not Meeting Work Requirements Will Reduce Low-Income Families' Access to Care and Worsen Health Outcomes on JSTOR.* Retrieved 2025-04-17, from `https://www.jstor.org/stable/resrep23819?seq=1`

Matta, S., Chatterjee, P., & Venkataramani, A. S. (2024, February). Changes in Health Care Workers' Economic Outcomes Following Medicaid Expansion. *JAMA*, *331*(8), 687–695. Retrieved 2025-04-17, from `https://doi.org/10.1001/jama.2023.27014` doi: 10.1001/jama.2023.27014

McGee, A. R., Akah, H., & Wiselogel, N. (2025). *The future of Group VIII (expansion) Medicaid coverage in Ohio | Publications.* Retrieved 2025-04-17, from `https://www.healthpolicyohio.org/our-work/publications/the-future-of-group-viii-expansion-medicaid-coverage-in-ohio`

Skinner, D. (2024, December). *Ohio poised to try Medicaid work requirements again • Ohio Capital Journal.* Retrieved 2025-04-17, from `https://ohiocapitaljournal.com/2024/12/19/ohio-poised-to-try-medicaid-work-requirements-again/`

Sommers, B. D., Chen, L., Blendon, R. J., Orav, E. J., & Epstein, A. M. (2020, September). Consequences of Work Requirements in Arkansas: Two-Year Impacts on Coverage, Employment, and Affordability of Care. *Health affairs (Project Hope)*, *39*(9), 1522–1530. Retrieved 2025-04-17, from `https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7497731/` doi: 10.1377/hlthaff.2020.00538

Wier, L. M. (2011). All-Cause Readmissions by Payer and Age, 2008.

Wing, C., Simon, K., & Bello-Gomez, R. A. (2018, April). Designing Difference in Difference Studies: Best Practices for Public Health Policy Research. *Annual Review of Public Health*, *39*(Volume 39, 2018), 453–469. Retrieved 2025-04-17, from `https://www.annualreviews.org/content/journals/10.1146/annurev-publhealth-040617-013507` (Publisher: Annual Reviews) doi: 10.1146/annurev-publhealth-040617-013507

Zuckerman, R. B., Sheingold, S. H., Orav, E. J., Ruhter, J., & Epstein, A. M. (2016, April). Readmissions, Observation, and the Hospital Readmissions Reduction Program. *New England Journal of Medicine*, *374*(16), 1543–1551. Retrieved 2025-04-17, from `https://www.nejm.org/doi/full/10.1056/NEJMsa1513024` (Publisher: Massachusetts Medical Society _eprint: https://www.nejm.org/doi/pdf/10.1056/NEJMsa1513024) doi: 10.1056/NEJMsa1513024